

Sravan Lavudya

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EDUCATION

Master of Science (MS), Mineral and Energy Economics Colorado School of Mines, CO, USA	Aug 2024 – May 2026
Bachelor of Technology in Mining and Mineral Engineering Ballarpur Institute of Technology, Chandrapur, Maharashtra, India	Nov 2020 – July 2023
Diploma in Mining and Mineral Engineering State Board of Technical Education and Training, Mancherial, Telangana, India	June 2017 - Oct 2020

TECHNICAL SKILLS

Mining and Geotechnical: Open-Pit Mining, Underground Mining, Rock Mechanics, Slope Stability Analysis, Highwall Inspections, Geological Mapping, Core Logging, Block model manipulation (Datamine, Surpac), Drilling Coordination.

Engineering Design: AutoCAD, MineSight, Deswik, Datamine, Surpac, Leapfrog Geo, 3D Geological Modeling.

Technical Analysis: Resource Estimation, Mine Ventilation, Field Data Collection, Engineering Calculations, Technical Reporting, Python, R, SQL, Excel.

RESEARCH EXPERIENCE

Research Assistant Payne Institute for Public Policy Golden, CO August 2025 – July 2026	1-Year
- Conducted research on mineral markets and trade dependencies, particularly between the U.S., China, and Africa. - Built financial models and performed complex engineering calculations and technical data analysis in Python and R to support feasibility studies, production economics, and risk assessment. - Collaborated with industry and government stakeholders to evaluate IRA-driven project economics for circular mineral supply chains and align research outputs with policy and investment priorities. - Analyzed critical mineral and geothermal supply-chain data to identify deployment bottlenecks, cost drivers, and operational risks affecting resource development projects. - Used SQL to query and extract high-volume mineral market datasets, improving data preparation workflows for policy, market, and economic analysis. - Delivered clear, well-organized written deliverables and technical memos, distilling complex engineering and economic findings into concise recommendations that improved decision-making across project teams.	

PROFESSIONAL EXPERIENCE

Management Trainee Sri Balaji Groups Karimnagar, Telangana, India August 2023 – July 2024	1-Year
- Authored and delivered 2 JORC/NACRI/IMIC-compliant mineral resource and reserve reports, providing precise data that successfully cleared internal audits and regulatory reviews. - Managed short-term engineering design and bench readiness across 2+ active extraction benches at a high-volume open-pit copper mine, ensuring safe operations and zero timeline delays. - Manipulated 3D geological block models using Datamine Studio and Leapfrog Geo to support resource evaluation and mine planning. - Coordinated a 3-rig exploration drilling program, executing precise geological mapping and 1,200+ meters of geotechnical core logging to strengthen subsurface characterization. - Solved open-pit versus sub-level caving methods for a 5-Lakh tonne per annum (LTPA) project, identifying technical changes that optimized extraction efficiency and saved ₹1.2 Crore in development costs. - Worked with the maintenance crew to eliminate 40+ weekly hours of equipment idle time, successfully reducing fuel waste and securing a consistent mechanical availability baseline of 22 out of 24 operational hours per day - Conducted daily highwall and bench inspections across 5+ active benches, successfully identifying 5+ critical rockfall hazards and ground instabilities to protect a daily field crew of 8 miners. - Used Deswik and AutoCAD to build precise 3D engineering models for underground infrastructure, saving 6 hours per week on technical reporting workflows and accelerating engineering calculations. - Optimized short-range production planning for a 1.5M tonne per annum (MTPA) footprint by aligning bench readiness and haulage access, consistently hitting 98% of daily extraction targets. - Coordinated maintenance, production, and haulage crews consisting of 60+ personnel to resolve scheduling conflicts, reducing shift handover delays by 20 minutes per day	

INTERNSHIP EXPERIENCE

Graduate Engineer Trainee Singareni Collieries Company Limited (SCCL) Telangana, India December 2022 – May 2023	6 months
Continuous Surface Miner: Sustainable Coal Production in India - Evaluated Continuous Surface Miner performance, equipment sizing, and mining methods to eliminate bottlenecks, reducing cycle-time delays by 10% and maximizing production throughput. - Audited and updated 3D block models in Surpac to refine resource estimation and enhance geological reporting accuracy for a 2-million-tonne ore body. - Tracked daily production output and equipment utilization data, optimizing haulage coordination to save 15+ operational minutes per shift during shift-level planning transitions. - Generated data-driven performance insights and equipment sizing models for engineering feasibility studies, directly guiding long-range planning decisions for new mining method viability. - Executed routine underground mine ventilation surveys and air quality monitoring across 3+ districts using anemometers, velometers, and digital gas detectors; maintained 100% compliance with Coal Mines Regulations (CMR) statutory safety limits for CH₄, CO, and CO₂.	
Diploma Industrial Trainee Singareni Collieries Company Limited (SCCL) Telangana, India December 2019 – May 2020	6 months
Open Pit Mine Design and Planning: - Executed 4+ comprehensive open-pit and underground mine design projects, managing pit optimization and haul road geometry to improve extraction sequencing and boost operational throughput. - Designed waste dumps and stockpiles for a 2.5M tonne per annum (MTPA) footprint, implementing precise slope stability analysis and safe lift sequencing to eliminate structural failure risks. - Conducted financial trade-off studies and capital cost estimations (OPEX/CAPEX) using cash-flow modeling to maximize project NPV during the engineering planning phase. - Evaluated environmental impacts of mineral extraction workflows, optimizing resource recovery strategies to reduce waste handling costs by ₹75 Lakh while securing critical materials. - Generated a full suite of mine design deliverables including precise pit layouts, haul road alignments, and dump configurations utilizing MineSight, AutoCAD, and Deswik to drive short- and long-term production schedules.	

SELECTED PUBLICATIONS

Payne Institute Student Commentary | April 2025 – July 2026

1. [Waste Not: Reimagining Mine Liabilities as Energy Assets](#); 2. [The Functioning and Trade of Minerals and Metals on Exchanges vs. Chemical Markets](#). 3. [How Europe’s Carbon Tax is Reshaping Mineral Markets](#); 4. [The Stabilizing Force of Aquifer Thermal Energy Storage in the 2026 Energy Landscape](#).

Research focus: critical minerals, mine liabilities, geothermal energy, mineral markets, carbon policy, and energy transition strategy.